Trainings

General Information

⚠ CAUTION ⚠

Do not use diesel engine oil in a natural gas engine. If diesel engine oil is used, valve torching, piston scuffing, and reduction of spark plug life will occur.

Cummins® natural gas engines require special oil that is available from major oil suppliers. Careful attention **must** be paid to engine oil specifications, because natural gas engine oil has different properties than diesel engine oil.

The use of correct engine lubricating oils, combined with appropriate oil drain and lubricating oil filter change intervals, is a critical factor in maintaining engine performance and durability. A sharp increase in component wear and damage can occur if lubricating oil and filter maintenance intervals are **not** followed as outlined in the maintenance schedule.

Cummins Inc. recommends the use of a high-quality multi-viscosity heavy-duty engine lubricating oil that meets the requirements of Cummins® Engineering Standard (CES) for natural gas engines. For further details and an explanation of engine lubricating oils for Cummins® engines, see Fluids for Cummins® Products Service Manual, Bulletin 5411406.

Shortened drain intervals can be required with single-grade oils, as determined by close monitoring of the oil condition with scheduled oil sampling. Use of single-grade oils can affect engine oil control.

Synthetic engine oils, American Petroleum Institute (API) Group III and Group IV base stocks, are recommended for use in Cummins® engines operating in ambient temperature conditions consistently below -25°C [-13°F]. Synthetic 0W-30 oils that meet API Group III and Group IV base stocks can be used in operations where the ambient temperature never exceeds 0°C [32°F]. Higher cylinder wear can be experienced when using 0W-30 oils in high-load situations.

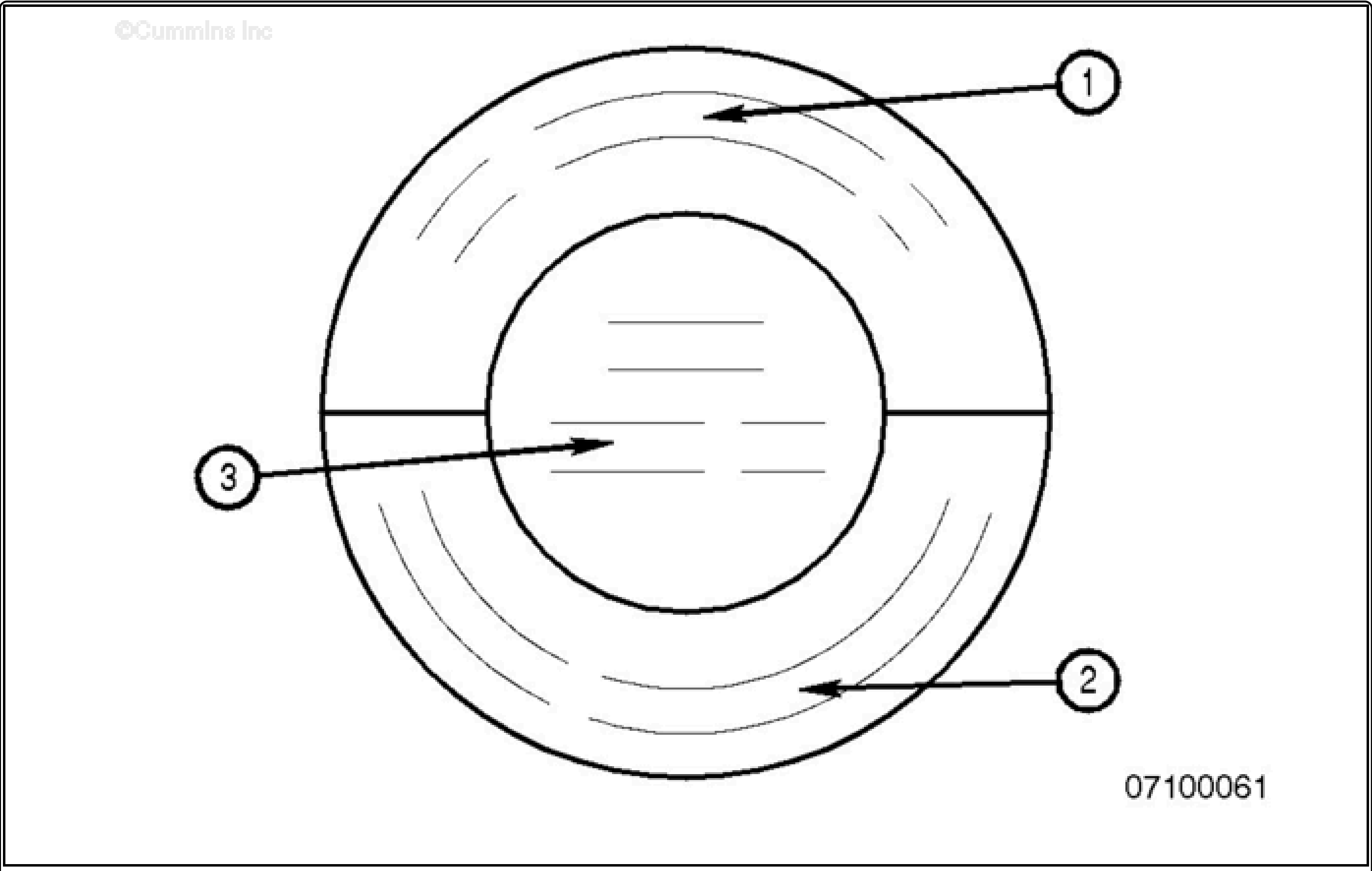
As the engine oil becomes contaminated, essential oil additives are depleted. Lubricating oils protect the engine while these additives are functioning properly.

Progressive contamination of the oil between oil and filter change intervals is normal. The amount of contamination will vary, depending on the operation of the engine, time on the oil, fuel consumed, and new oil added.

Note : Oil for natural gas engines does not darken and look dirty like diesel oil does. Use the maintenance schedule to determine the oil change interval requirement, not the oil's appearance.

Additional information regarding lubricating oil availability throughout the world is available in the Engine Manufacturing Association Lubricating Oils Data Book for Heavy Duty Automotive and Industrial Engines. The data book can be ordered from:

- Engine Manufacturers Association
- 2 North LaSalle Street
- Suite 2200
- Chicago, IL, U.S.A. 60602
- Phone: (312) 827-8700
- Facsimile: (312) 827-8737
- www.enginemanufacturers.org



The API service symbols are shown in the accompanying illustration.

1. The upper half of the symbols displays the appropriate oil categories.
2. The lower half contains words to describe additional oil information.
3. The center section identifies the Society of Automotive Engineers (SAE) oil viscosity grade.

Oil viscosity **must** be chosen according to the typical climate conditions experienced by the user. Use of 10W-30 is recommended for the best engine durability at higher ambient temperatures. For cold conditions, 5W-30 viscosity can be used for easier starting, improved oil flow, and improved fuel economy.

AfterMarket Oil Additive Usage

Cummins Inc. does **not** recommend the use of aftermarket oil additives. High-quality engine lubricating oils are very sophisticated, with precise amounts of additives blended into the lubricating oil to meet stringent requirements defined in CES.

These furnished oils meet performance characteristics that conform to the lubricant industry standards. Aftermarket lubricating oil additives are **not** necessary to enhance engine oil performance and in some cases can reduce the finished oil's capability to protect the engine.

New Engine Break-in Oils

Special break-in engine lubricating oils are **not** recommended for new or rebuilt Cummins® engines. Use the same lubricating oil that will be used during normal operation.

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